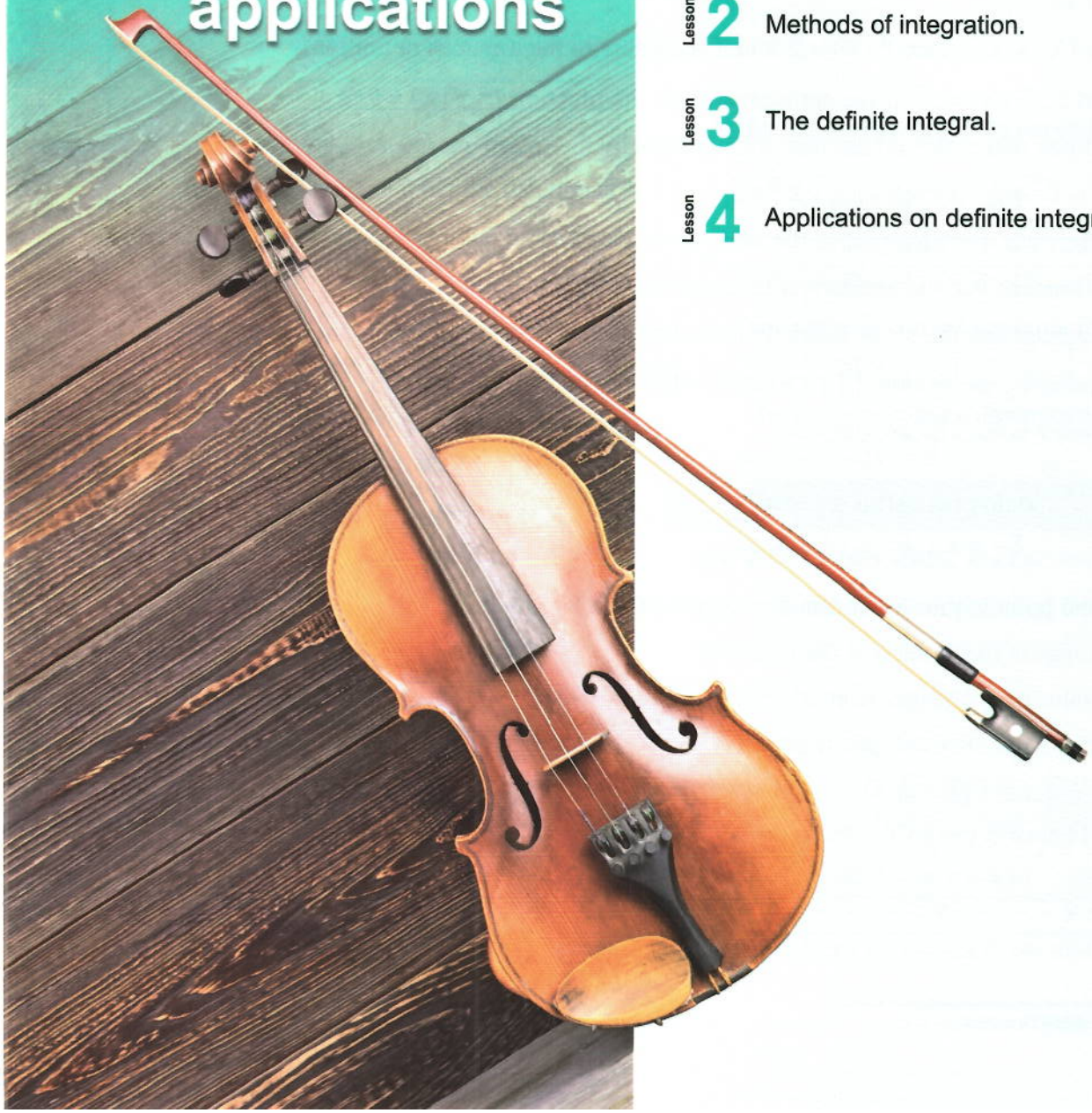
UNIT

THREE

The definite integral and its applications



⊙ Pre-requirements for unit 3

Introduction for integral – differential.

Lesson 1 Integration involves exponential and logarithmic functions.

Lesson 2 Methods of integration.

Lesson 3 The definite integral.

Lesson 4 Applications on definite integral.

Pre-requirements for Unit 3
Introduction for integral - differential

Introduction for integration

Definition

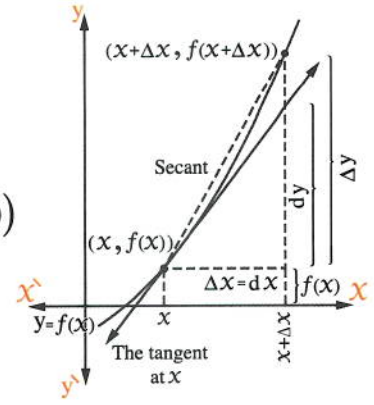
It is said that the function F is antiderivative to the function f
, if $\grave{F}(x) = f(x)$ for each x in the domain of f

For example :

If $F(x) = x^2$, then $\grave{F}(x) = 2x$ and according to the previous definition ,
then F : $F(x) = x^2$ is the antiderivative to the function f : $f(x) = 2x$
But you notice that all the functions defined by the following rules :
x^2 , $x^2 + 3$, $x^2 - 5$, ... , $x^2 + c$
(where c is a constant) have the same derivative $2x$
This means that the antiderivative to the function $2x$ is not unique.
Its agreed that the set of antiderivatives has been called the indefinite integration and denoted by the symbol $[\int f(x)\, dx]$, then : $\int f(x)\, dx = F(x) + c$
where c is an arbitrary constant.

Differential of the function y = f (x)

If $y = f(x)$ is a differentiable function
at the point whose x-coordinate = x and $(x + \Delta x)$
belongs to the domain of the function.
If value of x changes from x to $x + \Delta x$
, then value of y changes from y to $y + \Delta y$ $\left(\text{where } y = f(x)\right)$
, $y + \Delta y = f(x + \Delta x)$
and hence $\Delta y = f(x + \Delta x) - f(x)$
, $\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$
, from the definition of the derivative we know :



$$\grave{f}(x) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$\therefore$ When $\Delta x \longrightarrow 0$, then $\frac{\Delta y}{\Delta x} \longrightarrow \grave{f}(x)$

i.e. $\frac{\Delta y}{\Delta x} \simeq \grave{f}(x)$ when $\Delta x \longrightarrow 0$, $\Delta x \neq 0$

$\therefore \Delta y \simeq \grave{f}(x)\, \Delta x$

Definition

Let function f be differentiable on an open interval containing x and Δx is denoted for the change in x where $\Delta x \neq 0$, then

1. The differential of y (is denoted by the symbol d y) $= \grave{f}(x)\, \Delta x$

2. The differential of x (is denoted by the symbol d x) $= \Delta x$

i.e. d y = $\boxed{\grave{f}(x)\, dx}$ or $\boxed{dy = \grave{y}\, dx}$

For example :

If $y = x^3$, then $dy = 3x^2\, dx$

, If $z = \ell^4$, then $dz = 4\ell^3\, d\ell$

Remark

To calculate the change on the function $y = f(x)$ when value of x changes from x to $x + \Delta x$ we find $\Delta y = f(x + \Delta x) - f(x)$ or use the differential of the function **i.e.** $dy = \grave{f}(x)\, dx$, then we obtain an approximating value of change of the function and the difference between Δy and dy is decreased as Δx is decreased.

Example 1

Find the differential of each of the following :

1. $y = \frac{x}{x+1}$

2. $y = (2x - 3)^5$

3. $y = e^{2x-3}$

4. $A = 4\pi r^2$

Solution

1. $\because y = \frac{x}{x+1}$

$\therefore \grave{y} = \frac{(x+1) - x}{(x+1)^2} = \frac{1}{(x+1)^2}$

, $\because dy = \grave{y}\, dx$

$\therefore dy = \frac{1}{(x+1)^2}\, dx$

2. $\because y = (2x - 3)^5$ $\therefore \grave{y} = 5(2x-3)^4 \times 2 = 10(2x-3)^4$

, $\because dy = \grave{y}\, dx$ $\therefore dy = 10(2x-3)^4\, dx$

3. $\because y = e^{2x-3}$ $\therefore \grave{y} = 2e^{2x-3}$

, $\because dy = \grave{y}\, dx$ $\therefore dy = 2e^{2x-3}\, dx$

4. $\because A = 4\pi r^2$ $\therefore \grave{A} = 8\pi r$

, $\because dA = \grave{A}\, dr$ $\therefore dA = 8\pi r\, dr$

Example 2

If each of y and z are differentiable function with respect to x , then prove that :

1. $d(y \,.\, z) = y\, dz + z\, dy$ **2.** $d\left(\frac{y}{z}\right) = \frac{z\, dy - y\, dz}{z^2}$ (where $z \neq 0$)

Solution

1. $d(yz) = (yz)\grave{}\, dx = (y\grave{z} + z\grave{y})\, dx$

$= y\grave{z}\, dx + z\grave{y}\, dx$

$= y(\grave{z}\, dx) + z(\grave{y}\, dx)$

$= y\, dz + z\, dy$

Notice that

$dz = \grave{z}\, dx$

, $dy = \grave{y}\, dx$

2. $d\left(\frac{y}{z}\right) = \left(\frac{y}{z}\right)\grave{}\, dx = \frac{z\grave{y} - y\grave{z}}{z^2}\, dx$

$= \frac{z\grave{y}\, dx - y\grave{z}\, dx}{z^2} = \frac{z(\grave{y}\, dx) - y(\grave{z}\, dx)}{z^2} = \frac{z\, dy - y\, dz}{z^2}$

Unit 3

Lesson One

Integration involves exponential and logarithmic functions

Integration of exponential function :

- $\int e^{x}\, dx = e^{x} + c$
- $\int e^{kx+a}\, dx = \frac{e^{kx+d}}{k} + c$ where $k \neq 0$
- If $f(x)$ is differentiable function , then :

  $\int \hat{f}(x)\, e^{f(x)}\, dx = e^{f(x)} + c$ where c is an arbitrary constant.

Example 1

Find :

1. $\int 9\, e^{x}\, dx$
2. $\int e^{6x}\, dx$
3. $\int (4x^{3} - 4\, e^{8x})\, dx$
4. $\int (\sin 3x + 2\, e^{-4x})\, dx$
5. $\int (3\, e^{\frac{5}{4}y} - 8\, e^{-0.2y})\, dy$
6. $\int (e^{z} + e^{-z})^{2}\, dz$
7. $\int \frac{e^{x}+6}{e^{x}}\, dx$

Solution

1. $\int 9\, e^{x}\, dx = 9 \int e^{x}\, dx = 9\, e^{x} + c$

2. $\int e^{6x}\, dx = \frac{e^{6x}}{6} + c$

3. $\int (4x^{3} - 4\, e^{8x})\, dx = x^{4} - 4 \times \frac{e^{8x}}{8} + C = x^{4} - \frac{1}{2}\, e^{8x} + c$

4. $\int (\sin 3x + 2\, e^{-4x})\, dx = \frac{-\cos 3x}{3} + 2 \times \frac{e^{-4x}}{-4} + c$

   $= -\frac{1}{3} \cos 3x - \frac{1}{2}\, e^{-4x} + c$

5. $\int (3\, e^{\frac{5}{4}y} - 8\, e^{-0.2y})\, dy = \frac{3 \times e^{\frac{5}{4}y}}{\frac{5}{4}} - 8 \times \frac{e^{-0.2y}}{-0.2} + c$

   $= \frac{12}{5}\, e^{\frac{5}{4}y} + 40\, e^{-0.2y} + c$

6. $\int (e^{z} + e^{-z})^2 \, dz = \int (e^{2z} + 2 + e^{-2z}) \, dz$

$$= \frac{e^{2z}}{2} + 2z + \frac{e^{-2z}}{-2} + c$$

$$= \frac{1}{2} e^{2z} + 2z - \frac{1}{2} e^{-2z} + c$$

7. $\int \left(\frac{e^{x} + 6}{e^{x}}\right) dx = \int (1 + 6e^{-x}) \, dx = x + \frac{6e^{-x}}{-1} + c = x - 6e^{-x} + c$

Example 2

Find each of the following :

1. $\int x^3 e^{x^4} \, dx$

2. $\int (x - 1) e^{3x^2 - 6x + 5} \, dx$

3. $\int \sin x \times e^{\cos x} \, dx$

4. $\int \sec^2 x \, e^{\tan x} \, dx$

Solution

1. Put $f(x) = x^4$ $\quad \therefore \grave{f}(x) = 4x^3$

$\therefore \int x^3 e^{x^4} \, dx = \frac{1}{4} \int (4x^3) e^{x^4} \, dx = \frac{1}{4} e^{x^4} + c$

2. Put $f(x) = 3x^2 - 6x + 5$, $\quad \therefore \grave{f}(x) = 6x - 6 = 6(x - 1)$

$\therefore \int (x - 1) e^{3x^2 - 6x + 5} \, dx = \frac{1}{6} \int 6(x - 1) e^{3x^2 - 6x + 5} \, dx = \frac{1}{6} e^{3x^2 - 6x + 5} + c$

3. Put $f(x) = \cos x$ $\quad \therefore \grave{f}(x) = -\sin x$

$\therefore \int \sin x \times e^{\cos x} \, dx = -\int (-\sin x) \times e^{\cos x} \, dx = -e^{\cos x} + c$

4. $f(x) = \tan x$ $\quad \therefore \grave{f}(x) = \sec^2 x$

$\therefore \int \sec^2 x \, e^{\tan x} \, dx = e^{\tan x} + c$

Enrich your knowledge

$\because a^x = e^{\ln a^x} = e^{x \ln a}$

$\therefore \int a^x \, dx = \int e^{x \ln a} \, dx = \frac{e^{x \ln a}}{\ln a} + c = \frac{a^x}{\ln a} + c$

i.e. $\boxed{\int a^x \, dx = \frac{a^x}{\ln a} + c}$

Example 3

Find each of the following :

1. $\int 3^x \, dx$

2. $\int (x^{5e} + e^{5x} + 5^x) \, dx$

Solution

1. $\int 3^x \, dx = \frac{3^x}{\ln 3} + c$

2. $\int (x^{5e} + e^{5x} + 5^x) \, dx = \frac{x^{5e+1}}{5e+1} + \frac{e^{5x}}{5} + \frac{5^x}{\ln 5} + c$

Integration involves logarithmic function

$\because \frac{d}{dx}[\ln|x|] = \frac{1}{x}$, where $x \neq 0$

$\therefore \int \frac{1}{x} \, dx = \ln|x| + c$, where $x \neq 0$

Generally : If f is differentiable function , $f(x) \neq 0$

, then : $\int \frac{f'(x)}{f(x)} \, dx = \ln|f(x)| + c$, where c is an arbitrary constant.

Example 4

Find each of the following :

1. $\int \frac{3}{x} \, dx$

2. $\int \frac{9}{x \ln 2} \, dx$

3. $\int \frac{\ln x^4}{x \ln x^5} \, dx$

4. $\int \frac{3e}{x} + \frac{x}{3e} \, dx$

5. $\int \frac{x^2 - 25}{x^2 - 5x} \, dx$

6. $\int \frac{x+2}{x^2 + 4x + 3} \, dx$

Solution

1. $\int \frac{3}{x} \, dx = 3 \int \frac{1}{x} \, dx = 3 \ln|x| + c$

2. $\int \frac{9}{x \ln 2} \, dx = \frac{9}{\ln 2} \int \frac{1}{x} \, dx = \frac{9}{\ln 2} \times \ln|x| + c = 9 \log_2|x| + c$

3. $\int \frac{\ln x^4}{x \ln x^5} \, dx = \int \frac{4 \ln x}{5x \ln x} \, dx = \frac{4}{5} \int \frac{1}{x} \, dx$

$= \frac{4}{5} \ln|x| + c$

Remember

$\frac{\ln a}{\ln b} = \log_b a$

Remark

$\log_b a \times \log_a c = \log_b c$

4. $\int \left(\frac{3\,e}{x} + \frac{x}{3\,e}\right) dx = 3\,e \ln|x| + \frac{1}{6\,e}x^2 + c$

5. $\int \frac{x^2 - 25}{x^2 - 5x} dx = \int \frac{(x+5)(x-5)}{x(x-5)} dx = \int \frac{x+5}{x} dx = \int \left(1 + \frac{5}{x}\right) dx = x + 5 \ln|x| + c$

6. $\because f(x) = x^2 + 4x + 3 \;,\; \acute{f}(x) = 2x + 4 = 2(x+2)$

$\therefore \int \frac{x+2}{x^2+4x+3} dx = \frac{1}{2}\int \frac{2(x+2)}{x^2+4x+3} dx = \frac{1}{2}\ln|x^2 + 4x + 3| + c$

Remarks

1. $\int \tan x\, dx = \int \frac{\sin x}{\cos x} dx = -\int \frac{-\sin x}{\cos x} dx$

$= -\ln|\cos x| + c = \ln|\sec x| + c$

Note that
The numerator is the derivative of the denominator.

2. $\int \cot x\, dx = \int \frac{\cos x}{\sin x} dx$

$= \ln|\sin x| + c = -\ln|\csc x| + c$

Note that
The numerator is the derivative of the denominator.

3. $\int \sec x\, dx = \int \frac{\sec x(\sec x + \tan x)}{(\sec x + \tan x)} dx$

(by multiplying numerator and denominator by $(\sec x + \tan x)$)

$= \int \frac{\sec^2 x + \sec x \cdot \tan x}{\sec x + \tan x} dx$

$= \ln|\sec x + \tan x| + c$

Note that
The numerator is the derivative of the denominator.

4. $\int \csc x\, dx = \int \frac{\csc x(\csc x + \cot x)}{(\csc x + \cot x)} dx$

(by multiplying numerator and denominator by $(\csc x + \cot x)$)

$= -\int \frac{-\csc^2 x - \csc x \cot x}{\csc x + \cot x} dx$

$= -\ln|\csc x + \cot x| + c$

Note that
The numerator is the derivative of the denominator.

Example 5

Find each of the following :

1. $\int \cot(2x-5)\,dx$ **2.** $\int \frac{\sin 2x}{1+\sin^2 x}\,dx$ **3.** $\int \csc(3+4x)\,dx$

Solution

1. $\int \cot(2x-5)\,dx = \int \frac{\cos(2x-5)}{\sin(2x-5)}\,dx$

$= \frac{1}{2}\int \frac{2\cos(2x-5)}{\sin(2x-5)}\,dx$ (Note the numerator is the derivative of the denominator)

$= \frac{1}{2}\ln|\sin(2x-5)| + c$

2. $\int \frac{\sin 2x}{1+\sin^2 x}\,dx = \int \frac{2\sin x\cos x}{1+\sin^2 x}\,dx$ (Note the numerator is the derivative of the denominator)

$= \ln|1+\sin^2 x| + c$

3. $\int \csc(3+4x)\,dx = \int \frac{\csc(3+4x)\,[\csc(3+4x)+\cot(3+4x)]}{[\csc(3+4x)+\cot(3+4x)]}\,dx$

$= -\frac{1}{4}\int \frac{-4\csc^2(3+4x) - 4\csc(3+4x)\cot(3+4x)}{\csc(3+4x)+\cot(3+4x)}\,dx$

(Note the numerator is the derivative of the denominator)

$= -\frac{1}{4}\ln|\csc(3+4x)+\cot(3+4x)| + c$

Example 6

If the slope of the tangent to the curve of the function f at any point on it (x, y), equals $\frac{5x+3}{x}$, and the curve passes through the point $(e, 5e+7)$, find the equation of the curve.

Solution

$\because \frac{dy}{dx} = \frac{5x+3}{x}$

$\therefore \int \frac{dy}{dx}\,dx = \int \frac{5x+3}{x}\,dx = \int \left(5+\frac{3}{x}\right)dx$

$\therefore y = 5x + 3\ln|x| + c$

$\because$ The curve passes through $(e, 5e+7)$

$\therefore 5e+7 = 5e + 3\ln e + c$ $\quad\quad$ $\therefore c = 4$

$\therefore$ Equation of the curve is $y = 5x + 3\ln|x| + 4$

Example 7

If the rate of change of area of a lamina is a (in square centimeters) , with respect to t (in seconds) by the relation $\frac{d\,a}{d\,t} = e^{-0.2\,t}$, if the area of the lamina at the begining of changing is 140 cm^2. , then find area of the lamina after $\frac{1}{3}$ minutes.

Solution

$\frac{d\,a}{d\,t} = e^{-0.2\,t}$

$\therefore \int \frac{d\,a}{d\,t}\, d\,t = \int e^{-0.2\,t}\, d\,t$

$\therefore a = -5\, e^{-0.2\,t} + c$

at the begining of changing $t = 0$, $a = 140$

$\therefore c = 145$

$\therefore a = -5\, e^{-0.2\,t} + 145$

, after $\frac{1}{3}$ minutes = 20 seconds

$\therefore a = -5\, e^{-4} + 145$ cm^2

Example 8

Find each of the following :

1. $\int \frac{\cos x - \sin x}{\cos x + \sin x}\, d\,x$

2. $\int \frac{6}{x \ln x^3}\, d\,x$

3. $\int \frac{6}{x} (\ln x)^5\, d\,x$

4. $\int \frac{\sqrt{5 + \ln x}}{x}\, d\,x$

5. $\int \left(\frac{\log x}{x}\right) d\,x$

Solution

1. Put $f(x) = \cos x + \sin x$

$\therefore \grave{f}(x) = -\sin x + \cos x = \cos x - \sin x$

$\therefore \int \frac{\cos x - \sin x}{\cos x + \sin x}\, d\,x = \ln|\cos x + \sin x| + c$

2. $\int \frac{6}{x \ln x^3}\, d\,x = \int \frac{6}{3\,x \ln x}\, d\,x = 2\int \frac{1}{x \ln x}\, d\,x$

Put $f(x) = \ln x$ $\therefore \grave{f}(x) = \frac{1}{x}$

$\therefore$ The integration $= 2\int \frac{\frac{1}{x}}{\ln x}\, d\,x = 2 \ln|\ln x| + c$

3. Put $f(x) = \ln x$ $\quad\quad$ $\therefore \grave{f}(x) = \frac{1}{x}$

$$\therefore \int \frac{6}{x} (\text{lin } x)^5 \, dx = 6 \int \frac{1}{x} (\ln x)^5 \, dx = \frac{6 (\ln x)^6}{6} + c = (\ln x)^6 + c$$

4. Put $f(x) = 5 + \ln x$ $\quad\quad$ $\therefore \grave{f}(x) = \frac{1}{x}$

$$\therefore \int \frac{\sqrt{5 + \ln x}}{x} \, dx = \int \frac{1}{x} (5 + \ln x)^{\frac{1}{2}} \, dx$$

$$= \frac{2}{3} (5 + \ln x)^{\frac{3}{2}} + c$$

5. $\because \log x = \frac{\ln x}{\ln 10}$

$$\therefore \int \frac{\log x}{x} \, dx = \int \frac{\ln x}{x \ln 10} \, dx = \int \frac{\ln x}{x \ln 10} \, dx$$

$$= \frac{1}{\ln 10} \int \frac{\ln x}{x} \, dx = \frac{1}{\ln 10} \times \frac{1}{2} (\ln x)^2 + c$$

$$= \frac{1}{2 \ln 10} (\ln x)^2 + c$$